# Effat University

## Lab 3: Requirement Engineering

**SPRING 2026 – CS3151**

**ENGINEERING COLLEGE – COMPUTER SCIENCE DEPARTMENT**


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| COURSE NAME: Software Engineering | COURSE NUMBER: CS3151 | SECTION: 1-2-3 |


# Transforming User Stories to Requirements

## BodyBalance AI Case Study

### Learning Objectives

By completing this lab, you will be able to:

1. Identify and extract the three key components of a user story: Role (Actor), Feature (Function), and Benefit (Object)
2. Transform informal user stories into formal, SMART requirements
3. Apply the SMART criteria (Specific, Measurable, Achievable, Relevant, Time-bound) to requirements
4. Recognize how a single user story can expand into multiple detailed requirements

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## Part 1: Understanding the BodyBalance AI Domain

BodyBalance AI is a health, fitness, and recovery decision-support platform designed to help users analyze their physical and mental condition, assess posture deviations, and receive personalized activity or recovery guidance.

**1.1 Key Actors:**

- Active Student / Fitness User: Provides daily reports on one's physical and psychological state, registers pain zones, uploads posture photos, and follows customized recommendations for activities/recoveries.
- Fitness Coach / Personal Trainer: Views fitness and recovery profiles of clients to adjust their training programs safely.
- Healthcare Professional (Physiotherapist / Clinician): Reviews authorized symptom history and recovery timeline of patients along with decision-support reports produced by AI.
- System (BodyBalance AI / BodyScope AI Engine): Analyzes posture images, extracts posture information, generates personalized recommendations, and performs safety and red-flag checks.

**1.2 Key Entities:**

- User Account & Profile: Personal information, physical parameters, fitness experience, and user credentials.
- Physical and Mental Check-In: Daily records for exhaustion level, stress level, sleeping time, energy level, and physical activity levels.
- Pain & Posture Report: User inputs of body-map location on interactive body map, pain level ratings (1-10), and image files for posture check.
- Recommendation & Recovery Plan: Recommendations of activities for user based on system analysis.
- Escalation Trigger: A detected red-flag condition that stops routine exercise recommendations and directs the user toward appropriate professional/clinical care.
- Permission Entity: User-controlled permissions to manage data access.

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## Part 2: User Story Template and Structure

A well-formed user story follows this structure:

```
As an Active student/Fitness user,
I want to complete a daily physical and mental check-in,
so that the system can understand my current condition before providing exercise guidance.
```

**Key Components:**

| Component | Definition | Example |
|---|---|---|
| Role (Actor) | Who is performing the action? | Active student / Fitness user, Fitness coach, Healthcare Professional |
| Feature (Function) | What action or capability is needed? | Complete daily check-in, select pain zones on body map, upload posture photo |
| Benefit (Object) | Why is this needed? What value does it provide? | Understand condition before exercise, track recovery progress, detect posture asymmetry |

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## Part 3: User Stories from BodyBalance AI Platform Examples

Below are user stories extracted from the BodyBalance AI case study. Study each one and identify the Role, Feature, and Benefit.

### User Story 1: Pain Location Identification

```
As an Active student/Fitness User,
I want to identify the specific location of my pain on an interactive body map,
so that I can accurately report where I am experiencing discomfort.
```

Role (Actor): Active Student/ Fitness User
Feature (Function): Identify my pain location on an interactive body map

Benefit (Object): Accurately report where I am experiencing physical discomfort

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### User Story 2: Pain Intensity Recording

```
As an Active student/Fitness User,
I want to record the intensity of my pain for each selected pain area,
so that BodyBalance AI can consider the severity of my discomfort when providing recovery guidance.
```

Role (Actor): Active Student/ Fitness User
Feature (Function): Record the intensity of my pain for each selected pain area

Benefit (Object): Help BodyBalance AI consider the severity of my discomfort when providing recovery guidance

### User Story 3: BodyScope AI Posture Scan

```
As an Active student/Fitness User,
I want to upload a posture photo for BodyScope AI analysis,
so that I can receive objective feedback on my joint alignment and balance.
```

Role (Actor): Active student/Fitness User
Feature (Function): Upload posture photos for automated computer vision analysis
Benefit (Object): Identify posture alignment and asymmetry indicators through automated analysis.

User Story 4: Tailored Exercise Recommendations

```
As an Active student/Fitness User,
I want to receive personalized exercise recommendations,
so that I can train safely without exacerbating existing soreness.
```

Role (Actor): Active Student / Fitness User
Feature (Function): Receive personalized exercise routines based on current physical status
Benefit (Object): Exercise safely at an appropriate intensity without aggravating pain

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## User Story 5: Personalized Recovery Plan

```
As an Active student/Fitness User,
I want to receive a customized recovery plan,
so that I can follow appropriate rest and stretching protocols.
```

Role (Actor): Active Student / Fitness User

Feature (Function): Receive customized rest and mobility plans
Benefit (Object): Support physical recovery and appropriate rest and mobility

## User Story 6: Red-Flag Clinical Escalation

```
As an Active student/Fitness User,
I want to be alerted immediately when concerning symptoms are detected,
so that I know when to seek medical evaluation.
```

Role (Actor): Active Student / Fitness User
Feature (Function): Receive an immediate safety warning when predefined red-flag symptoms are detected.
Benefit (Object): Know when professional or clinical evaluation may be required instead of continuing with routine self-care recommendations.

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## User Story 7: Long-Term Progress Tracking

```
As an Active student/Fitness User,
I want to view historical charts of my pain and mobility metrics,
so that I can track my overall recovery progress.
```

Role (Actor): Active Student / Fitness User
Feature (Function): View historical trends of pain levels, check-ins, posture results, and recovery information.

Benefit (Object): Monitor health improvement and recovery trends over time

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## User Story 8: Professional Referral Summary

```
As an Active student/Fitness User,
I want to generate a downloadable health summary,
so that I can share accurate symptom records with my healthcare provider.
```

Role (Actor): Active Student / Fitness User
Feature (Function): Export a downloadable summary report of symptoms, posture scans, and check-in history
Benefit (Object): Share organized symptom, posture, and recovery information with an authorized healthcare professional.

## User Story 9: Coach Review & Training Adjustments

```
As a Fitness coach,
I want to access authorized client recovery records,
so that I can adjust workout programs according to the client's current condition and recovery status.
```

Role (Actor): Fitness Coach / Personal Trainer
Feature (Function): Review authorized client fitness, recovery, and fatigue data
Benefit (Object): Adjust training intensity and activities according to the client's current condition and recovery status.

### User Story 10: Data Sharing & Privacy Management

```
As a User,
I want to manage my data-sharing permissions,
so that my personal health information is accessible only to authorized professionals.
```

Role (Actor): Registered User
Feature (Function): Control permissions for coach and clinician access to health logs
Benefit (Object): Protect health privacy and ensure personal data is shared only with authorized individuals

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## Part 5: Exercises

### Exercise 1: Extract Components from User Stories

For each user story below, identify and write down the Role (Actor), Feature (Function), and Benefit (Object):

User Story A:

```
As an active student/fitness user,
I want to record my physical limitations and past injuries in my profile,
so that the system avoids generating exercises that could aggravate my condition.
```

Your Answer:

- Role: Active Student / Fitness User
- Feature: Record physical limitations and past injuries in the user profile
- Benefit: Avoid generating exercise recommendations that could aggravate existing conditions

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User Story B:

```
As a fitness coach,
I want to receive automatic alerts when a client logs a high pain or fatigue score,
so that I can modify their upcoming training session before they arrive.
```

Your Answer:

- Role: Fitness Coach / Personal Trainer
- Feature: Receive automatic physical condition alerts for high pain or fatigue scores
- Benefit: Modify the client's upcoming training session according to their reported pain or fatigue level.

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User Story C:

```
As a healthcare professional,
```

```
I want to view an exported PDF summary of a patient's 30-day symptom and posture history,
```

```
so that I have organized data to support my diagnostic assessment.
```

Your Answer:

- Role: Healthcare Professional (Physiotherapist / Clinician)
- Feature: View an exported 30-day symptom and posture history summary report
- Benefit: Support professional clinical assessments with organized, objective patient data

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# Part 4: Transformation Examples

## Example 1: Daily Physical and Mental Check-In

Original User Story:

```
As an Active Student / Fitness User,
```

```
I want to complete a daily physical and mental check-in,
```

```
so that the system can understand my current condition before providing exercise or recovery guidance.
```

Extracted Components:

- Role: Active Student / Fitness User
- Feature: Complete a daily physical and mental check-in
- Benefit: Help the system understand the user's current condition and provide appropriate guidance

Transformed into SMART Requirements:

| Req ID | Requirement | SMART Criteria |
|---|---|---|
| REQ-1 | The system must allow an Active Student / Fitness User to record their current physical condition and mental well-being during a daily check-in. | S: Physical and mental condition recorded; M: Check-in information stored; A: Implemented through input fields; R: Supports personalized guidance; T: During each daily check-in |
| | **The system must allow the user to complete a daily physical and mental check-in before receiving exercise or recovery recommendations.** | |
| REQ-2 | The system must allow the user to record physical limitations, discomfort, or relevant previous injuries during the daily check-in. | S: Records limitations and injuries; M: Information stored; A: Form-based input; R: Supports safer recommendations; T: During check-in |
| | **The system must allow the user to record physical limitations, discomfort, and previous injuries during the daily check-in.** | |
| REQ-3 | The system must use the completed physical and mental check-in information when generating personalized exercise or recovery recommendations. | S: Check-in information used; M: Data included in recommendation process; A: Recommendation engine accesses stored data; R: Supports personalization; T: Before recommendations are generated |
| | **The system must use the user's completed physical and mental check-in information before generating personalized exercise or recovery recommendations.** | |

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## Example 2: Pain Assessment and Mapping

Original User Story:

```
As an Active Student / Fitness User,,
```

```
I want to identify where I am experiencing pain and record its intensity,
```

```
so that BodyBalance AI can consider my current discomfort when providing exercise and recovery guidance.
```

Extracted Components:

- Role: Active Student / Fitness User
- Feature: Identify pain locations and record pain intensity
- Benefit: Help the system understand current discomfort and provide appropriate guidance

Transformed into SMART Requirements:

| Req ID | Requirement | SMART Criteria |
|---|---|---|
| REQ-4 | The system must provide an interactive body map that allows the user to identify up to 3 distinct pain zones during each assessment. | S: Interactive body map with maximum 3 zones; M: Number of zones recorded; A: Body-map interface; R: Identifies pain locations; T: During each assessment |
| | **The system must allow the user to select up to 3 distinct pain zones using the interactive body map during each assessment.** | |
| REQ-5 | The system must allow the user to record pain intensity for each selected pain zone using a scale from 1 to 10. | S: 1–10 pain scale; M: Pain value recorded; A: Input validation; R: Standardizes pain information; T: During assessment |
| | **The system must allow the user to record the intensity of each selected pain zone using a 1–10 scale.** | |
| REQ-6 | The system must store the selected pain zones and corresponding pain intensity values as part of the user's assessment record. | S: Pain information stored; M: Zones and intensity saved; A: Database storage; R: Supports recommendations and progress tracking; T: After assessment |
| | **The system must store the user's selected pain zones and pain intensity values in the assessment record after submission.** | |

---

## Example 3: BodyScope AI Posture Analysis Transformation

Original User Story:

```
As an Active Student / Fitness User,
I want to upload a posture image for BodyScope AI analysis,
so that the system can provide information about my posture and alignment.
```

Extracted Components:

- Role: Active Student / Fitness User
- Feature: Upload a posture image and receive BodyScope AI analysis
- Benefit: Understand posture and alignment information

Transformed into SMART Requirements:

| Req ID | Requirement | SMART Criteria |
|---|---|---|
| REQ-7 | The system must allow users to upload posture images in JPEG or PNG format for BodyScope AI analysis. | S: JPEG/PNG images; M: File format validated; A: Image upload functionality; R: Supports posture analysis; T: During upload |
| REQ-8 | The system must reject posture image files larger than 10 MB. | S: 10 MB maximum; M: File size validated; A: Upload validation; R: Controls supported image size; T: Before processing |
| REQ-9 | The system must validate that an uploaded posture image has adequate lighting and sufficient full-body visibility within 3 seconds. | S: Lighting and visibility validation; M: Validation completed; A: Automated image validation; R: Improves analysis quality; T: Within 3 seconds |
| REQ-10 | BodyScope AI must extract 33 pose landmarks from a valid posture image. | S: 33 pose landmarks; M: Landmark count measured; A: BodyScope AI processing; R: Supports posture analysis; T: During processing |
| REQ-11 | BodyScope AI must calculate shoulder inclination and pelvic tilt in degrees from a valid posture image. | S: Two posture measurements; M: Measurements provided in degrees; A: Landmark-based calculation; R: Supports posture information; T: During analysis |
| REQ-12 | The system must complete BodyScope AI posture processing within 5 seconds for a valid uploaded image. | S: Posture processing; M: Completion time measured; A: Defined processing target; R: Provides timely results; T: Within 5 seconds |

---

## Example 4: Personalized Exercise Recommendations Transformation

Original User Story:

```
As an Active Student / Fitness User,
```

```
I want the system to provide personalized exercise recommendations based on my current physical condition,
```

```
so that I can follow activities that are appropriate for my condition and existing pain.
```

Extracted Components:

- Role: Active Student / Fitness User
- Feature: Receive personalized exercise recommendations
- Benefit: Follow activities appropriate for the user's current physical condition

Transformed into SMART Requirements:

| Req ID | Requirement | SMART Criteria |
|---|---|---|
| REQ-13 | The system must use the user's recorded pain locations and pain intensity when generating personalized exercise recommendations. | S: Use pain location and intensity; M: Pain information included in recommendation process; A: Recommendation logic; R: Supports personalized exercise guidance; T: Before recommendations are generated |
| REQ-14 | The system must consider the user's physical and mental check-in information when generating personalized exercise recommendations. | S: Consider check-in information; M: Check-in data used; A: System data integration; R: Supports condition-aware recommendations; T: Before recommendations are generated |
| REQ-15 | The system must consider the user's available BodyScope AI posture assessment results when generating personalized exercise recommendations. | S: Use posture assessment results; M: Posture information included; A: BodyScope AI integration; R: Supports personalized recommendations; T: Before recommendations are generated |
| REQ-16 | The system must check the user's reported symptoms against the 8 predefined red-flag clinical indicators before generating a routine exercise plan. | S: Check 8 predefined red flags; M: All indicators checked; A: Automated safety check; R: Supports safer exercise guidance; T: Before exercise-plan generation |
| REQ-17 | The system must prevent routine exercise-plan generation when a predefined red-flag clinical indicator is detected. | S: Block routine plan; M: Exercise plan is not generated; A: System workflow control; R: Supports safety escalation; T: Immediately after red-flag detection |
| REQ-18 | The system must display a safety warning and appropriate professional or clinical referral information when a red flag is detected. | S: Warning and referral information displayed; M: Information is visible to the user; A: Predefined safety and referral content; R: Supports appropriate professional care; T: Immediately after detection |

---

## Exercise 2: Transform User Stories to SMART Requirements

Transform the following user story into at least 5 SMART requirements. Use the format provided:

User Story:

```
As an Active Student / Fitness User,
I want to track my pain and recovery history,
so that I can monitor changes in my condition over time.
```

Your Transformed Requirements:

| Req ID | Requirement | SMART Criteria |
|---|---|---|
| REQ-1 | The system must allow users to record their pain level during each follow-up assessment. | S: Record pain level; M: Pain level uses a 1–10 scale; A: Input field and validation; R: Supports pain and recovery tracking; T: During each follow-up assessment |
| | **The system must allow users to record their pain level using a 1–10 scale during each follow-up assessment.** | |
| REQ-2 | The system must save each follow-up assessment with its date and time. | S: Save assessment and timestamp; M: Date and time recorded; A: Database storage; R: Supports historical progress tracking; T: When the assessment is submitted |
| | **The system must save each completed follow-up assessment with its date and time when the user submits it.** | |
| REQ-3 | The system must allow users to review their previous pain levels and recovery information. | S: Display previous pain and recovery information; M: Previous records can be viewed; A: History interface; R: Helps users monitor progress; T: When the user accesses their history |
| | **The system must allow users to review their previous pain levels and recovery information whenever they access their progress history.** | |
| REQ-4 | The system must display the user's pain and posture assessment history in chronological order when the user accesses the progress history. | S: Display previous pain levels by date; M: Pain records and dates are displayed; A: History interface; R: Supports monitoring of changes over time; T: When the user accesses progress history. |
| | **The system must allow users to review their recorded pain levels organized by assessment date whenever they access their progress history.** | |
| REQ-5 | The system must display repeated posture assessment results in the user's progress history. | S: Display posture assessment results; M: Previous posture results can be viewed; A: Stored BodyScope AI results; R: Supports recovery and posture monitoring; T: When the user accesses progress history. |
| | **The system must display the user's previous BodyScope AI posture assessment results whenever the user accesses their progress history.** | |

## Exercise 3: Identify Missing Requirements

The following user story has been transformed into requirements, but some important requirements are missing. Identify what is missing and add 2-3 additional requirements:

Original User Story:

```
As an Active Student / Fitness User,
I want to upload a posture image for BodyScope AI analysis,
so that I can receive information about my posture and visible physical indicators.
```

Existing Requirements:

- REQ-1: The system must allow users to upload a posture image.
- REQ-2: The system must process the posture image using BodyScope AI.
- REQ-3: The system must display the BodyScope AI analysis results.

What is missing?

The existing requirements do not specify important details about image format, maximum file size, image quality, processing time, and the specific information BodyScope AI must extract.

The missing requirements include:

1. The system should specify the supported image formats.
2. The system should specify the maximum image size.
3. The system should validate adequate lighting and full-body visibility.
4. The system should specify the required BodyScope AI processing time.
5. The system should specify the posture information extracted from the image.

---

Your Additional Requirements:

| Req ID | Requirement |
|---|---|
| REQ-4 | The system must accept posture images in JPEG or PNG format with a maximum file size of 10 MB. |
| REQ-5 | The system must validate adequate lighting and full-body visibility within 3 seconds after the posture image is uploaded. |
| REQ-6 | BodyScope AI must extract 33 pose landmarks and calculate shoulder inclination and pelvic tilt in degrees within 5 seconds for each valid posture image. |

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## Exercise 4: Create Your Own User Stories (Project Based)

Based on the BodyBalnace AI case study, create  new user stories for your project. For each story, ensure it follows the proper format and includes clear Role, Feature, and Benefit.

User Story 1:

```
As an Active Student / Fitness User,
I want to receive a customized recovery plan,
so that I can follow appropriate rest and mobility guidance based on my current condition.
```

User Story 2:

```
As an Active Student / Fitness User,
I want to view my pain and posture progress over time,
so that I can monitor changes in my recovery and physical condition.
```

User Story 3:

```
As an authorized Physiotherapist,
I want to review and modify AI-recommended exercises before they are published,
so that professional oversight can be applied to the user's exercise plan.
```

---

# Part 4: Group Project

## Exercise 4. 1: Generate all user stories of your project

**US-01| Pain Location Mapping**

As an Active Student / Fitness User,

I want to mark my pain areas on an interactive body map,

so that I can provide precise pain locations for my assessment.

**US-02 | Interactive Pain Mapping**

As an Active Student / Fitness User,

I want to record my pain points and intensity,

so that the system can provide personalized recovery guidance.

**US-03 | BodyScope AI Posture Scan**

As an Active Student / Fitness User,

I want to upload a posture photo for BodyScope AI analysis,

so that I can receive information about my posture and alignment.

**US-04 | Tailored Exercise Recommendations**

As an Active Student / Fitness User,

I want to receive personalized exercise recommendations,

so that I can follow activities appropriate for my current condition.

**US-05 | Personalized Recovery Plan**

As an Active Student / Fitness User,

I want to receive a customized recovery plan,

so that I can follow appropriate rest and mobility guidance.

**US-06 | Red-Flag Clinical Escalation**

As an Active Student / Fitness User,

I want to be alerted when concerning symptoms are detected,

so that I know when professional or clinical evaluation may be appropriate.

**US-07 | Long-Term Progress Tracking**

As an Active Student / Fitness User,

I want to view my pain and posture progress over time,

so that I can monitor changes in my recovery and physical condition.

**US-08 | Professional Referral Summary**

As an Active Student / Fitness User,

I want to generate a summary of my relevant health and recovery information,

so that I can share organized information with an authorized healthcare professional.

**US-09 | Coach Review and Training Adjustments**

As a Fitness Coach / Personal Trainer,

I want to access authorized client fitness and recovery information,

so that I can adjust training according to the client's current condition.

**US-10 | Data Sharing and Privacy Management**

As a User,

I want to manage my data-sharing permissions,

so that my personal information is accessible only to authorized individuals.

**US-11 | Physiotherapist Review**

As an authorized Physiotherapist,

I want to review and modify AI-recommended exercises before they are published,

so that professional oversight can be applied to the exercise plan.

**US-12 | Daily Physical and Mental Check-In**

As an Active Student / Fitness User,
I want to report my daily physical and mental condition,
so that the system can consider my current state when providing personalized guidance.

**US-13 | Health and Injury Profile**

As an Active Student / Fitness User,
I want to record relevant health conditions, previous injuries, and physical limitations,
so that the system can provide recommendations appropriate to my individual condition.

**US-14 | Recovery Feedback**

As an Active Student / Fitness User,
I want to provide feedback about my pain and recovery after following a recommended plan,
so that the system can monitor my response and adjust future recommendations.

**US-15 | Healthcare Professional Review**

As an authorized Healthcare Professional,
I want to access the user's shared symptom, pain, posture, and recovery history,
so that I can review the user's progress and use the information to support professional assessment.

# Exercise 4.2: Transform to Requirements

| Req ID | Requirement |
|---|---|
| REQ -1 | The system shall allow users to select up to 3 distinct pain zones using the interactive body map. |
| REQ -2 | The system shall allow users to record a pain intensity from 1 to 10 for each selected pain zone. |
| REQ -3 | The system shall accept posture images in JPEG or PNG format, with a maximum file size of 10 MB. |
| REQ -4 | The system shall validate adequate lighting and full-body visibility within 3 seconds of posture image upload. |
| REQ -5 | BodyScope AI shall extract 33 pose landmarks from each valid posture image. |
| REQ -6 | BodyScope AI shall calculate shoulder inclination and pelvic tilt in degrees from the extracted posture data. |
| REQ -7 | BodyScope AI shall complete posture analysis within 5 seconds after receiving a valid image. |
| REQ -8 | The system shall check the user's symptoms against 8 predefined red-flag clinical indicators before generating a routine exercise plan. |
| REQ -9 | If a red-flag indicator is detected, the system shall block routine exercise-plan generation and display a safety warning with appropriate clinical referral information. |
| REQ -10 | The system shall allow an authorized physiotherapist to modify, replace, or disable AI-recommended exercises before the exercise plan is published. |

| | |
|---|---|
| REQ -11 | The system shall allow users to record their physical condition and mental state during each daily check-in. |
| REQ -12 | The system shall allow users to record relevant health conditions, previous injuries, and physical limitations in their profile. |
| REQ -13 | The system shall use the user's recorded pain locations and pain intensity when generating personalized exercise recommendations. |
| REQ -14 | The system shall consider the user's physical and mental check-in information when generating personalized exercise recommendations. |
| REQ -15 | The system shall consider available BodyScope AI posture assessment results when generating personalized exercise recommendations. |
| REQ -16 | The system shall generate a personalized recovery plan that includes appropriate rest, mobility, and activity guidance based on the user's current assessment. |

| | |
|---|---|
| **REQ -17** | The system shall save each completed pain, posture, and recovery assessment with its date and time. |
| **REQ -18** | The system shall display the user's previous pain levels, posture assessment results, and recovery information in chronological order when the user accesses progress history. |
| **REQ -19** | The system shall allow users to record follow-up pain intensity using a 1–10 scale after following a recommended recovery plan. |
| **REQ -20** | The system shall use the user's latest follow-up information when generating subsequent personalized recommendations. |
| **REQ -21** | The system shall allow users to generate a summary containing their relevant pain, posture, symptom, and recovery information for sharing with an authorized healthcare professional. |
| **REQ -22** | The system shall allow a Fitness Coach / Personal Trainer to access a client's fitness and recovery information only after the client grants permission. |
| **REQ -23** | The system shall allow an authorized Healthcare Professional to access the user's shared symptom history, pain records, posture assessments, and recovery progress. |
| **REQ -24** | The system shall allow users to grant or revoke data-sharing permission for each Fitness Coach or Healthcare Professional. |
| **REQ -25** | The system shall prevent a Fitness Coach or Healthcare Professional from accessing the user's shared information immediately after the user revokes permission. |

## Exercise 4.3: Review the requirements to be SMART.

| SMART Criterion | Review of BodyBalance AI Requirements |
|---|---|
| Specific | The requirements clearly describe the functions the system must perform, such as selecting up to 3 pain zones, processing posture images, and checking red-flag symptoms. |
| Measurable | The requirements contain measurable values such as a maximum of 3 pain zones, pain intensity from 1–10, a 10 MB image limit, 33 pose landmarks, 8 red-flag indicators, and processing within 3 or 5 seconds. |
| Achievable | The requirements are designed according to the planned scope and capabilities of the BodyBalance AI system, including BodyScope AI posture analysis and professional review. |
| Relevant | The requirements directly support the purpose of BodyBalance AI by helping users report discomfort, assess posture, receive appropriate guidance, monitor progress, and obtain professional oversight when needed. |
| Time-bound | Time limits are specified where appropriate, such as validating image quality within 3 seconds and completing BodyScope AI posture analysis within 5 seconds. |

# Part 6: Key Takeaways

## Why Transform User Stories to Requirements?

2 Clarity: User stories are informal and conversational; requirements are formal and precise
3 Implementation Guidance: Requirements provide specific details for developers
4 Testing: Requirements can be tested; user stories are too vague for testing
5 Traceability: Requirements can be tracked and verified
6 Completeness: Transformation often reveals missing functionality

## SMART Criteria Explained

| Criterion | Definition | Example |
|---|---|---|
| Specific | Clear and well-defined, not ambiguous | The system must allow the user to select up to 3 distinct pain zones. |
| Measurable | Can be quantified or verified | Pain intensity must be recorded using a 1–10 scale. |
| Achievable | Realistic and feasible within constraints | The system must accept JPEG or PNG images up to 10 MB. |
| Relevant | Aligned with business goals and user needs | The system must use pain information when generating personalized exercise recommendations. |
| Time-bound | Specifies when or how often | BodyScope AI must complete posture processing within 5 seconds. |

## Common Mistakes to Avoid

7 Too Vague: "The system should be user-friendly" → Too subjective
8 Too Technical: Jumping to implementation details instead of requirements
9 Missing Context: Not considering who, what, and why
10 Incomplete: Forgetting edge cases and error handling
11 Unmeasurable: Requirements that cannot be tested or verified

Functional Requirements (FR)

FR-1: The system shall allow users to create and maintain a personal profile containing relevant fitness, physical limitation, and previous injury information.

FR-2: The system shall allow users to complete a daily physical and mental check-in including stress, energy, sleep, fatigue, symptoms, and recent physical activity.

FR-3: The system shall allow users to identify pain locations using an interactive body map and record the pain intensity for each selected area.

FR-4: The system shall allow users to upload posture images for BodyScope AI analysis.

FR-5: The system shall analyze valid posture images and provide posture and alignment information.

FR-6: The system shall generate personalized activity recommendations using the user's current check-in, pain information, profile information, recent activity, and available BodyScope AI results.

FR-7: The system shall generate personalized recovery plans containing appropriate rest, mobility, recovery, and follow-up guidance.

FR-8: The system shall allow users to view their pain, posture, check-in, and recovery progress over time.

FR-9: The system shall allow users to provide follow-up feedback after completing a recommended activity or recovery action and use relevant feedback in future recommendations.

FR-10: The system shall perform predefined red-flag safety checks before providing routine exercise guidance and shall provide appropriate escalation or referral guidance when a red flag is detected.

FR-11: The system shall allow users to generate a summary of relevant symptom, pain, posture, check-in, and recovery information for sharing with an authorized healthcare professional.

FR-12: The system shall allow users to grant and revoke access permissions for Fitness Coaches and Healthcare Professionals.

FR-13: The system shall allow an authorized Fitness Coach / Personal Trainer to review permitted client fitness and recovery information.

FR-14: The system shall allow an authorized Healthcare Professional / Physiotherapist to review permitted symptom history, pain records, assessment information, BodyScope AI results, and recovery progress.

FR-15: The system shall allow an authorized Physiotherapist to review, modify, replace, or disable AI-recommended exercises before the exercise plan is published.

**Non-Functional Requirements (NFR)**

NFR-1 [Security & Privacy]: The system shall require authenticated access to protected user information and enforce active user permissions whenever a Fitness Coach or Healthcare Professional attempts to access user data.

NFR-2 [Performance]: The system shall provide visible processing feedback whenever an assessment, recommendation request, or BodyScope AI analysis is submitted and clearly indicate whether processing is pending, completed, or failed.

NFR-3 [Usability]: The system shall provide clear labels, instructions, validation messages, and error-recovery guidance for check-ins, pain mapping, BodyScope AI image uploads, progress tracking, follow-up feedback, and permission management.

NFR-4 [Reliability]: The system shall not mark a check-in, assessment, feedback submission, or profile update as successfully completed unless all required information has been stored successfully.

NFR-5 [Safety]: The system shall complete all applicable predefined safety and referral checks before presenting routine personalized exercise guidance and shall prioritize the defined escalation response when a red-flag condition is detected.

NFR-6 [Transparency]: The system shall clearly identify AI-generated BodyScope AI, activity, recovery, and referral outputs as decision-support information rather than a confirmed medical diagnosis.

AI assistance :
It was used for minor assistance with wording, and organization. The lab content and answers were finalized by the students.